

Shell, TUI, and AI Coding CLI

DS 5110: Big Data Systems
Summer 2026
Lecture 1b

Yue Cheng



UNIVERSITY
of VIRGINIA

Some material taken/derived from:

- Wisconsin CS 544 by Tyler Caraza-Harter.

@ 2026 released for use under a [CC BY-SA](https://creativecommons.org/licenses/by-sa/4.0/) license.

Learning objectives

- Setting up an EC2 VM instance via AWS Academy
- Navigate a Linux file system
- Operating within a Linux shell
- Install OpenCode or whatever AI coding CLI that suits you

A shell is an agent

A shell program

Loop:

```
print(SOME PROMPT)
cmd = get_input()
run(cmd)
```



If you can type it, you can automate it

A shell program

Loop:

```
print(SOME PROMPT)
cmd = get_input()
run(cmd)
```



Script.sh

Shell commands

...
...

You can run a shell inside of a shell

A terminal window titled "bash" with a window icon in the top-left corner and a zoom icon in the top-right corner. The terminal shows a zsh prompt "Yue ~ \$" followed by the command "echo hello world" and its output "hello world". Then, the command "pwd" is entered, resulting in the output "/Users/yue". Finally, the command "bash" is entered, which results in a new prompt "bash-3.2\$".

```
Yue ~ $ echo hello world
hello world
Yue ~ $ pwd
/Users/yue
Yue ~ $ bash

The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh
-s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
bash-3.2$
```

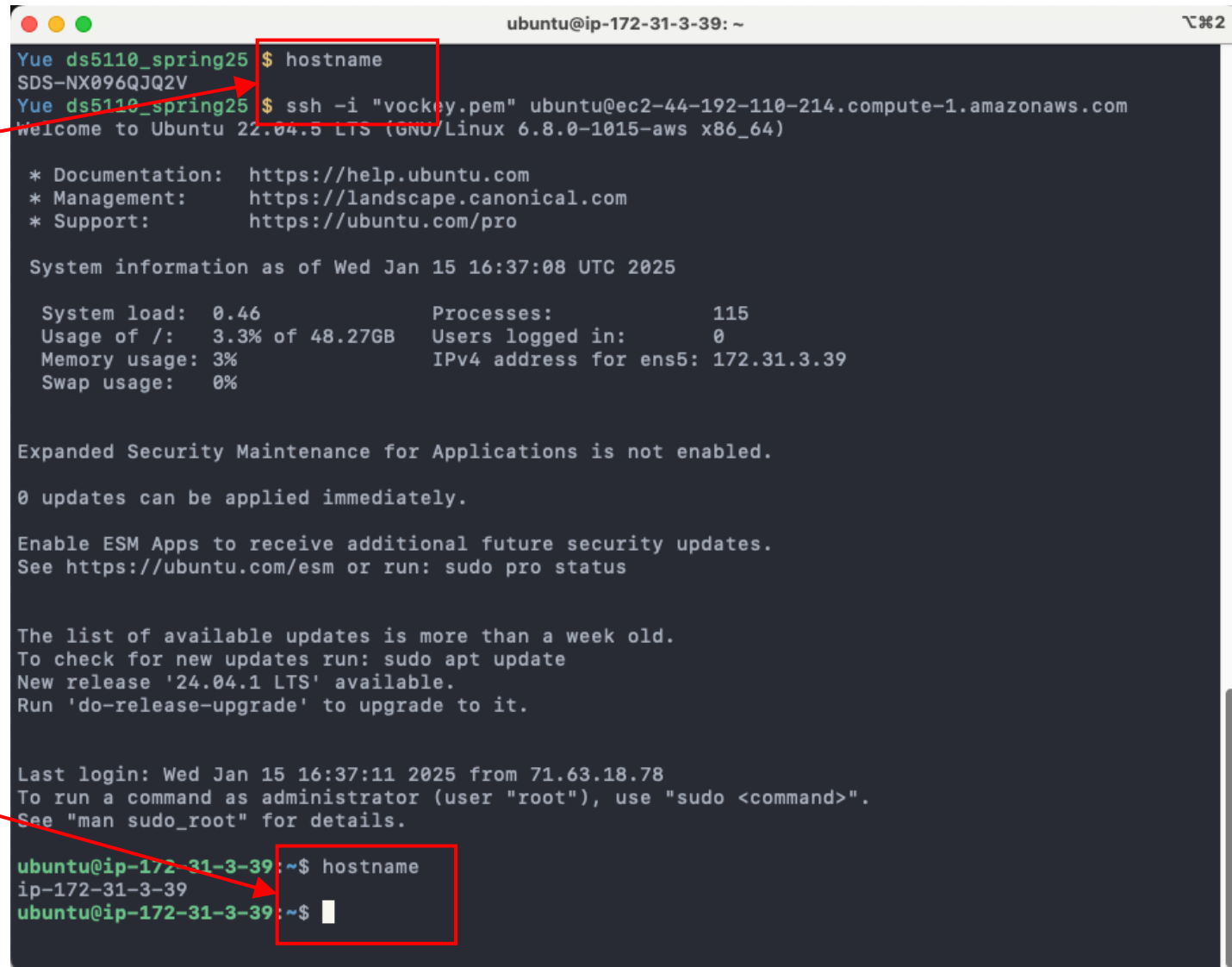
Annotations:

- zsh prompt: Points to the "Yue ~ \$" prompt.
- Start running bash: Points to the "bash" command.
- bash prompt: Points to the "bash-3.2\$" prompt.

SSH: Secure shell

Running on
my laptop

Running on
my EC2 VM



The image shows a terminal window titled 'ubuntu@ip-172-31-3-39: ~'. The user 'Yue ds5110_spring25' is at a prompt. They run 'hostname', which returns 'SDS-NX096QJQ2V'. Then they run 'ssh -i "vockey.pem" ubuntu@ec2-44-192-110-214.compute-1.amazonaws.com'. The terminal shows the SSH banner for Ubuntu 22.04.5 LTS, system information (load, processes, memory, etc.), and update notifications. At the bottom, the prompt changes to 'ubuntu@ip-172-31-3-39:~\$' after the user runs 'hostname' again, which returns 'ip-172-31-3-39'. Red boxes and arrows highlight the 'hostname' command and its output on both the laptop and the EC2 VM.

```
ubuntu@ip-172-31-3-39: ~
Yue ds5110_spring25 $ hostname
SDS-NX096QJQ2V
Yue ds5110_spring25 $ ssh -i "vockey.pem" ubuntu@ec2-44-192-110-214.compute-1.amazonaws.com
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jan 15 16:37:08 UTC 2025

System load:  0.46           Processes:            115
Usage of /:   3.3% of 48.27GB Users logged in:        0
Memory usage: 3%            IPv4 address for ens5: 172.31.3.39
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Wed Jan 15 16:37:11 2025 from 71.63.18.78
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-3-39:~$ hostname
ip-172-31-3-39
ubuntu@ip-172-31-3-39:~$
```

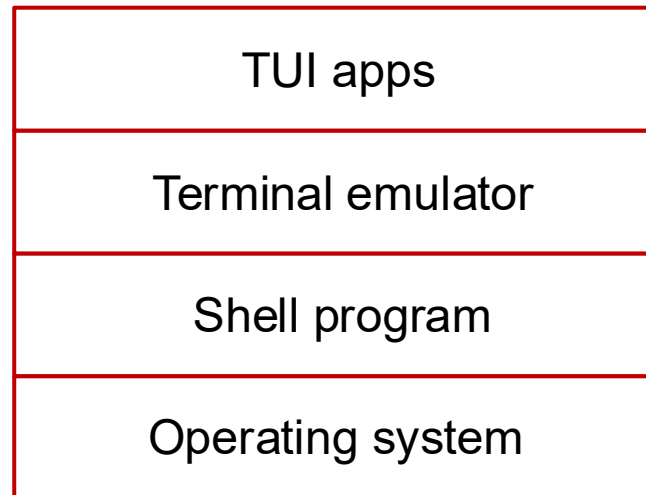
Terminal user interface (TUI)



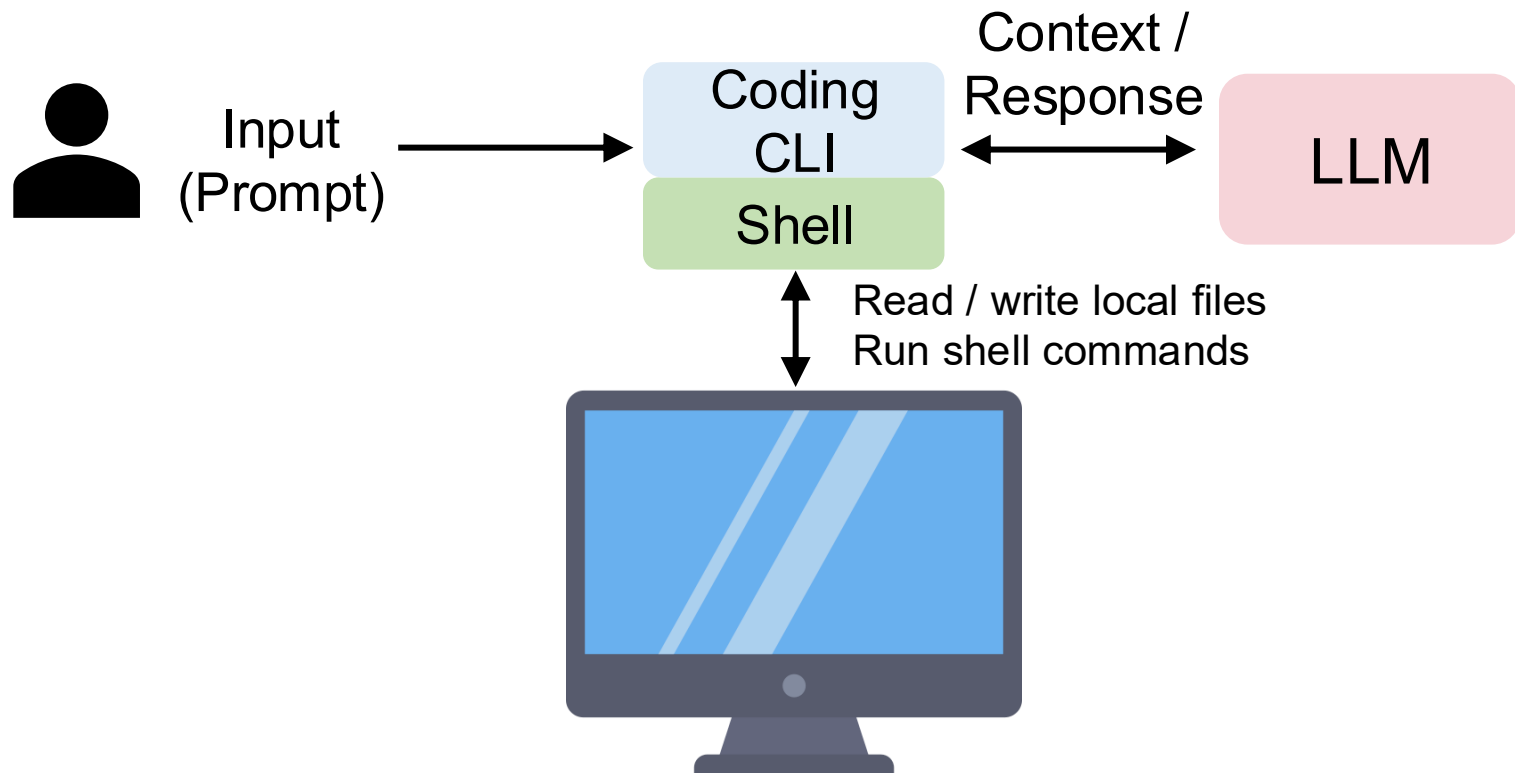
tmux



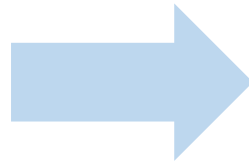
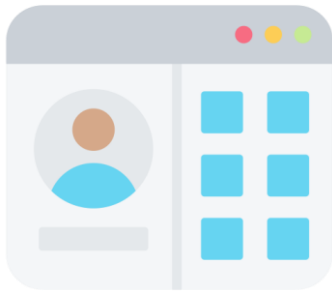
opencode



AI coding CLI (cmd-line interface)



AI agents as first-class users



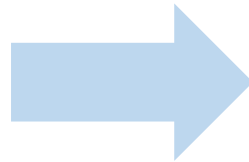
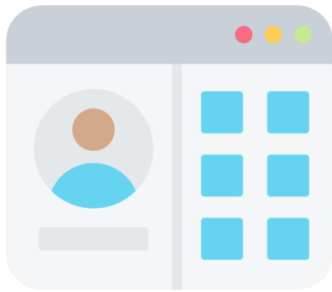
GUI (graphical user interface)
Designed for human users

CLI (command-line interface)
Designed for LLM agents

Visual feedback
Low cognitive load

Natural language / text driven
Global project context
Direct, efficient tool access

AI agents as first-class users



GUI (graphical user interface)
Designed for human users

CLI (command-line interface)
Designed for LLM agents

Visual feedback
Low cognitive load

Natural language / text driven
Global project context
Direct, efficient tool access

**How is this different from using
ChatGPT for coding?**

TODOs

- A0 is out
 - Due by 11:59pm ET, Friday, May 22
- Default CLI: OpenCode with free models
 - Codex CLI/Claude Code is also fine if you already have a paid subscription
- Sign up for AWS Academy
- Form a team of up to four students for assignments & projects
 - A team size of 3-4 is ideal
 - Working individually is also allowed, depending on the availability of OpenRouter credits
 - Each team will receive one OpenRouter API key
 - Watch for Canvas announcement for detailed instructions